Assignment 3: Github

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Assignment 3: Collaborating in Github

Introduction

This is the base R markdown file we can populate with our code via branching, then we can replace this section with the introduction and data description, as well as the planned plots/analyses.

Installing and Loading Packages

```
#Feel free to add any and all packages we might need
#install.packages("pastecs")
library(pastecs)
#install.packages("tidyverse")
library(tidyverse)
#install.packages("knitr")
library(knitr)
#install.packages("car")
library(car)
```

Data Preparation

```
#Reading in csv file and assigning to data frame object named "text_wide"
text_wide <- read.csv("TextMessages.csv")
class(text_wide$Group)
```

```
## [1] "integer"
```

```
text_wide$Group <-factor(text_wide$Group,levels=c(1,2))

#In case we need long format data-
text_long <- pivot_longer(
  text_wide,
  cols= c("Baseline","Six_months"),
  names_to = "Timepoint",
  values_to = "Text_count"
  )
class(text_long$Timepoint)
```

```
## [1] "character"
```

```
text_long$Timepoint<-factor(text_long$Timepoint)

#Moving participant number to the front
```

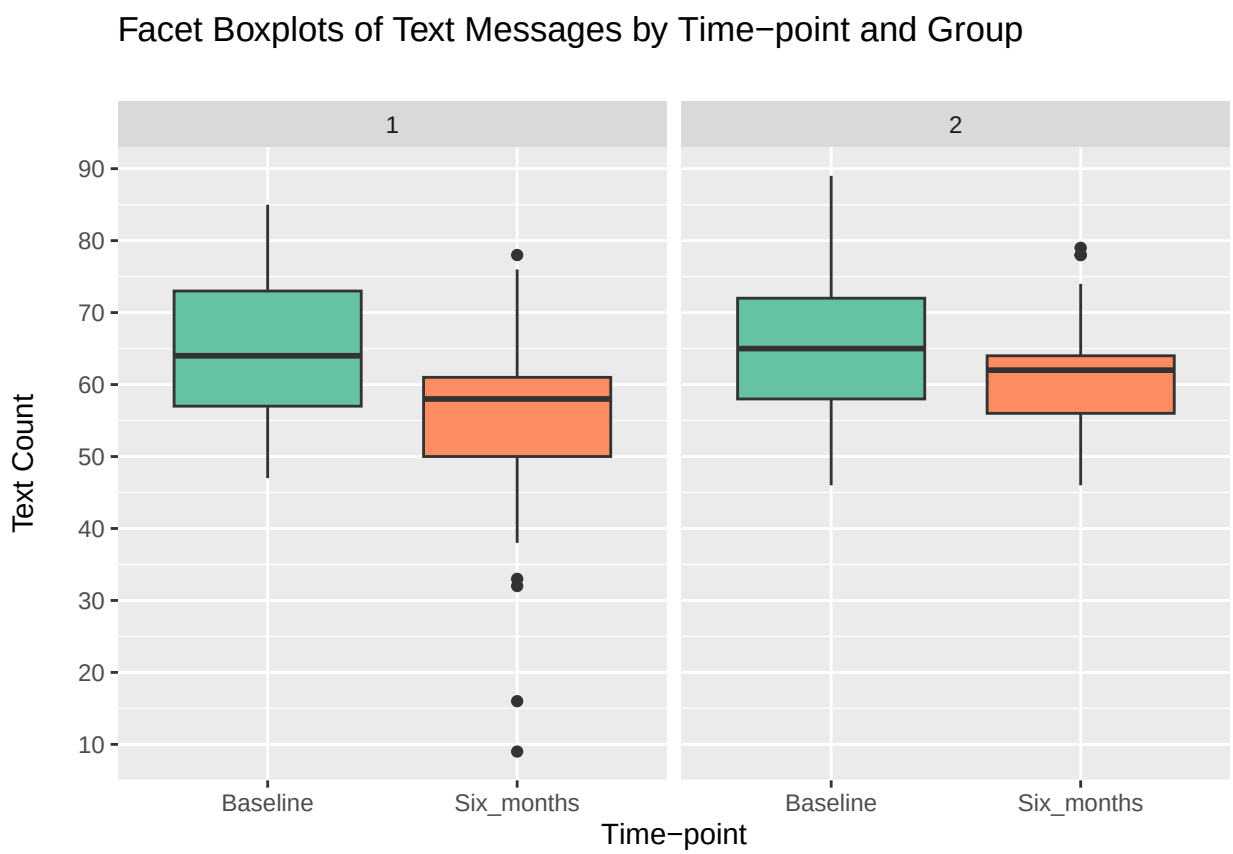
```
text_wide <- text_wide |> relocate("Participant")
text_long <- text_long |> relocate("Participant")
```

Visualization 1: Faceted Boxplot of text messages by Group and Time

```
#Creating faceted boxplots of text messages for each group and across the
#time-points using ggplot() and geom(boxplot). We color-code by time-points and
#use facet_wrap(~Group) to display the plots side by side.

visualization1<- ggplot(
  data=text_long,
  aes(y=Text_count,
      x=Timepoint,
      fill=Timepoint))+
  geom_boxplot(show.legend=FALSE)+
  facet_wrap(~Group)+
  scale_y_continuous(breaks=seq(from=0, to=100, by=10))+
  labs(title = "Facet Boxplots of Text Messages by Time-point and Group\n",
       y="Text Count\n",
       x="Time-point")+
  scale_fill_brewer(palette = "Set2")

print(visualization1)
```



```
#We can compare the distributions of text messages across time-points (Baseline,
#6-months) for both groups (1,2) side by side. At baseline, both groups show
#similar distributions of text message counts, with approximately the same
#median values of 64/65. The interquartile ranges and overall spread also looks
#comparable, which suggests that the two groups had similar baseline
#characteristics.

#At 6 months, we can see a decrease in median text message counts for both
#groups, though the reduction looks to be more for group 1 (from approximately
#64 to 58) when compared to group 2 (approximately 65 to 62). There are also
#noticeable differences in variability between the groups at six months. Group 1
#has a wider distribution, with several low outliers with values below 20
#messages as well as higher counts above 75. This suggests greater variability
#in text counts among participants in Group 1. In contrast, Group 2 has a
#narrower distribution with relatively higher messaging counts, suggesting more
#consistent messaging patterns across the participants.
```

Visualization 2: Faceted Bar Chart of text messages by Group and Time

```
#Code chunk for bar chart
```

Summary Statistics of Text Messages by Group and Time

```
#Code chunk for summary stats
```

Interpretation and conclusions

Any concluding comments in addition to individual interpretations above

Visualization 1 conclusions

Overall, the boxplots suggest a reduction in text messaging frequency in both groups over the six-month period, with a more noticeable shift in Group 1. Additionally, Group 1 demonstrates greater variability in messaging behavior at the six-month time-point, whereas Group 2 shows a more consistent distribution. The presence of several low outliers in Group 1 suggests that changes in messaging frequency may vary considerably among individual participants.